



United International University

QUEST FOR EXCELLENCE

Report: Caring Corner – Digital Old Age Home

Course Name: Project Management

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Section: G

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Chapter 1: Introduction

1.1 Purpose

The purpose of the **Caring Corner – Digital Old Age Home Management System** is to develop a centralized digital platform that simplifies and modernizes the management of elderly care services. The system is designed to eliminate manual processes, reduce communication gaps, and provide real-time access to critical information such as seat availability, admissions, and donations. By connecting family members, caregivers, old-age home managers, and administrators on a single platform, Caring Corner aims to improve transparency, operational efficiency, and quality of care for elderly individuals.

1.2 Problem Statement

1.2.1 Problem Background

Elderly care management in many old-age homes is still largely dependent on manual procedures, phone calls, and paper-based records. Families often face difficulties obtaining accurate information regarding bed availability, admission status, and care services. Donation processes also lack transparency, discouraging community participation. These inefficiencies result in poor coordination, delays, and mistrust among stakeholders.

1.2.2 Problem Description

Key problems identified include:

- Manual and inefficient seat booking and admission processes
- Lack of real-time visibility of bed availability
- Fragmented communication between families and care providers
- Limited transparency in donation collection and utilization
- Absence of structured feedback mechanisms

1.2.3 Problem Reasoning

These issues persist due to outdated management practices, lack of digital integration, and insufficient use of modern web technologies. Without a centralized system, data becomes inconsistent and difficult to manage, leading to operational inefficiencies and reduced quality of elderly care services.

Chapter 2: Motivation

2.1 Motivation

The motivation behind Caring Corner arises from the growing need for efficient, transparent, and accessible elderly care management. With increasing urbanization and busy lifestyles, families require reliable digital tools to manage elderly care remotely. This project is driven by the goal of improving the dignity, safety, and well-being of elderly individuals through technology.

2.2 Goals

- Simplify elderly care home discovery and booking
- Digitize admission and management processes
- Enable transparent and secure donation handling
- Improve communication between stakeholders
- Enhance overall care quality through better data management

Chapter 3: Vision & Scope

3.1 Vision

The vision of Caring Corner is to create a compassionate, transparent, and technology-driven ecosystem that ensures dignified and well-coordinated care for elderly individuals while empowering families and care providers with real-time information and efficient management tools.

3.2 Scope and Out of Scope

In-Scope:

- Online seat booking and admission management
- Real-time bed availability tracking
- Donation system with payment integration
- Feedback collection system
- Role-based admin dashboard

Out-of-Scope:

- Physical healthcare services
- Medical diagnosis or treatment systems
- Hardware-based monitoring solutions

Chapter 4: Methodology

4.1 What is Software Development Methodology?

A **software development methodology** is a structured framework that defines the processes, techniques, roles, and activities required to design, develop, test, deploy, and maintain a software system. It provides a systematic approach that guides a project from initial planning and requirement analysis through implementation and long-term maintenance. By following a defined methodology, development teams can ensure better coordination, improved quality control, efficient resource utilization, and timely delivery of the software product.

Software development methodologies help manage project complexity by dividing work into manageable phases, establishing clear responsibilities, and providing mechanisms for monitoring progress and handling risks. Commonly used methodologies include **Waterfall**, **Agile**, **Scrum**, **Spiral**, **Incremental**, and **Extreme Programming (XP)**. Each methodology has its own strengths

and limitations, making it suitable for different types of projects depending on factors such as project size, requirement stability, stakeholder involvement, and risk level.

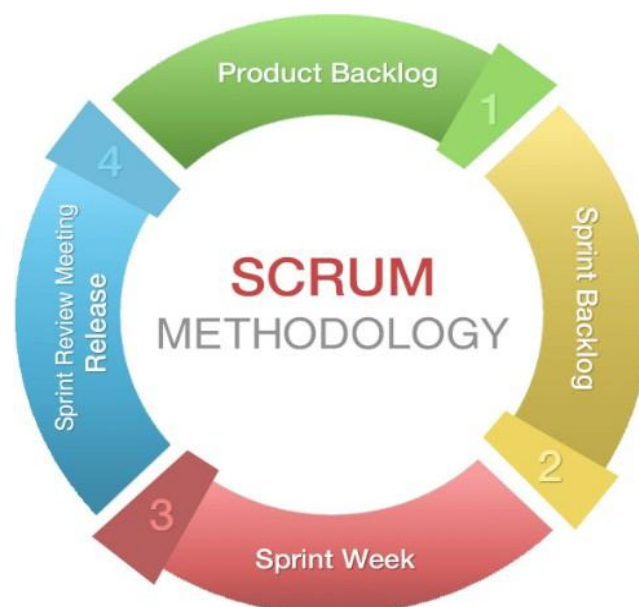
For a system like **Caring Corner**, which involves multiple stakeholders, evolving requirements, and user-centered features such as real-time seat availability, online booking, donation systems, and administrative dashboards, choosing the right methodology is critical to project success.

4.2 Methodology Selection

After evaluating several software development methodologies—including **Waterfall**, **Incremental**, **Spiral**, **Agile**, **Scrum**, and **Extreme Programming (XP)**—the project team selected the **Scrum methodology** for developing the Caring Corner system.

Traditional models like Waterfall were found to be less suitable because they require fixed requirements at the beginning of the project and offer limited flexibility for change. Given that Caring Corner is a digital platform that must adapt to user feedback, stakeholder input, and operational challenges of old-age home management, a rigid sequential model would increase the risk of rework and project delays.

Scrum, which is a framework under the Agile umbrella, was chosen because it supports **iterative development**, **continuous feedback**, and **incremental delivery**. This approach allows the project team to deliver functional components of the system in short development cycles called **sprints**, making it easier to respond to changes, improve system quality, and ensure stakeholder satisfaction throughout the development lifecycle.



4.3 Why Scrum Methodology for Caring Corner?

The Scrum methodology was selected for the Caring Corner project due to several key reasons that align closely with the nature and requirements of the system:

4.3.1 Evolving and Dynamic Requirements

Caring Corner serves multiple user groups, including family members, elderly users, old-age home managers, and administrators. Each group has different needs and expectations, which may evolve as the system is developed and tested. Scrum allows requirements to be refined continuously through sprint planning and backlog grooming, ensuring that the system remains aligned with real user needs without disrupting the overall project structure.

4.3.2 Incremental and Iterative Development

Scrum enables the development of the system in small, manageable increments. Core features such as online seat booking, admission management, real-time bed availability, and donation processing can be developed, tested, and delivered incrementally. This approach ensures that functional components are available early, reducing development risks and allowing stakeholders to evaluate progress regularly.

4.3.3 Faster Feedback and Continuous Improvement

Through regular **Sprint Reviews** and **Sprint Retrospectives**, the project team can gather feedback from stakeholders and identify areas for improvement. This feedback-driven development process helps reduce errors, improve usability, and enhance system performance before final deployment.

4.3.4 Improved Team Collaboration and Transparency

Scrum promotes strong collaboration through defined roles such as **Product Owner**, **Scrum Master**, and **Development Team**, along with regular meetings like **Daily Scrum** sessions. These practices improve communication, ensure transparency in task progress, and help identify and resolve issues early in the development cycle.

4.3.5 Risk Reduction and Quality Assurance

By implementing continuous testing and integration within each sprint, Scrum helps identify technical and operational risks early. For a system like Caring Corner—where payment integration, data security, and real-time information accuracy are critical—early risk detection significantly improves overall system reliability and quality.

4.3.6 Efficient Time and Resource Management

Scrum uses **time-boxed sprints**, which help the team stay focused and maintain a steady development pace. This structured time management reduces the likelihood of schedule overruns and ensures efficient use of available resources.

4.4 Scrum Roles in Caring Corner Project

The Scrum framework defines clear roles to ensure accountability and smooth project execution:

- **Product Owner:** Responsible for defining system requirements, prioritizing the product backlog, and ensuring that the developed features deliver maximum value to stakeholders.
- **Scrum Master:** Facilitates the Scrum process, removes obstacles, ensures adherence to Scrum practices, and supports effective team collaboration.

- **Development Team:** A cross-functional team responsible for designing, developing, testing, and delivering functional increments of the Caring Corner system.

4.5 Scrum Events and Artifacts

To manage the project effectively, the following Scrum events and artifacts are utilized:

- **Sprint Planning:** Determines sprint goals and selects backlog items for development.
- **Daily Scrum:** Short daily meetings to track progress and identify blockers.
- **Sprint Review:** Demonstrates completed features and gathers feedback.
- **Sprint Retrospective:** Evaluates team performance and identifies improvement areas.
- **Product Backlog:** A prioritized list of features, enhancements, and fixes.
- **Sprint Backlog:** Tasks selected for implementation during a sprint.

Chapter 5: Work Breakdown Structure (WBS)

5.1 What is Work Breakdown Structure?

A Work Breakdown Structure (WBS) is a hierarchical and deliverable-oriented decomposition of the total scope of a project into smaller, more manageable components called work packages. It provides a structured way to organize and define all the work required to successfully complete a project. The WBS begins with the overall project objective and progressively breaks it down into phases, deliverables, and individual tasks until each element becomes manageable, estimable, and assignable.

For the Caring Corner – Digital Old Age Home Management System, the WBS plays a critical role in ensuring that every functional and non-functional requirement is clearly identified, planned, and tracked. By using a WBS, the project team can systematically manage development activities such as requirement analysis, system design, development, testing, deployment, and maintenance. The WBS also serves as the foundation for project scheduling, cost estimation, resource allocation, and risk management.

The primary objectives of using a WBS in this project are to:

- Clearly define project deliverables
- Ensure complete coverage of project scope
- Assign responsibilities effectively
- Monitor progress and control changes
- Reduce project complexity through structured decomposition

5.2 WBS Design Principles

To ensure effectiveness and clarity, the WBS for the Caring Corner project follows standard WBS design principles:

5.2.1 The 100% Rule

The 100% Rule ensures that the WBS captures all the work required to complete the project—no more and no less. For Caring Corner, this means that every activity necessary for delivering features such as online seat booking, admission management, real-time bed availability, donation processing, feedback collection, and admin control is included. At the same time, any work that falls outside the approved project scope is excluded. The sum of all lower-level elements represents 100% of the project work.

5.2.2 Planned Outcomes, Not Planned Actions

The WBS is designed to be outcome-oriented rather than activity-focused. Instead of listing actions like “write code” or “test system,” the Caring Corner WBS emphasizes deliverables such as “Seat Booking Module,” “Donation System,” or “Admin Dashboard.” This approach ensures that every task directly contributes to tangible project results and supports objective progress tracking.

5.2.3 Level 2 is the Most Important

Level 2 elements of the WBS represent the major phases of the Caring Corner project. These include Project Initialization, Requirement Analysis, System Design, Development, Testing, Deployment, and Maintenance. Clearly defining these major components provides a strong structural backbone for project planning and simplifies coordination across the project lifecycle.

5.2.4 Four Elements in Each WBS Component

Each WBS element is defined with four essential attributes:

1. Scope of work – what is to be delivered
2. Schedule – the timeframe for completion
3. Responsibility – the person or team accountable
4. Effort estimation – the expected workload

This ensures that all work packages are measurable, time-bound, and properly assigned.

5.2.5 Mutually Exclusive Elements

All WBS elements in the Caring Corner project are mutually exclusive, meaning no task or deliverable overlaps with another. For example, the “Donation System” is treated separately from “Admission Management,” even though both involve payment-related logic. This separation avoids duplication, confusion, and accountability conflicts.

5.2.6 Appropriate Level of Decomposition

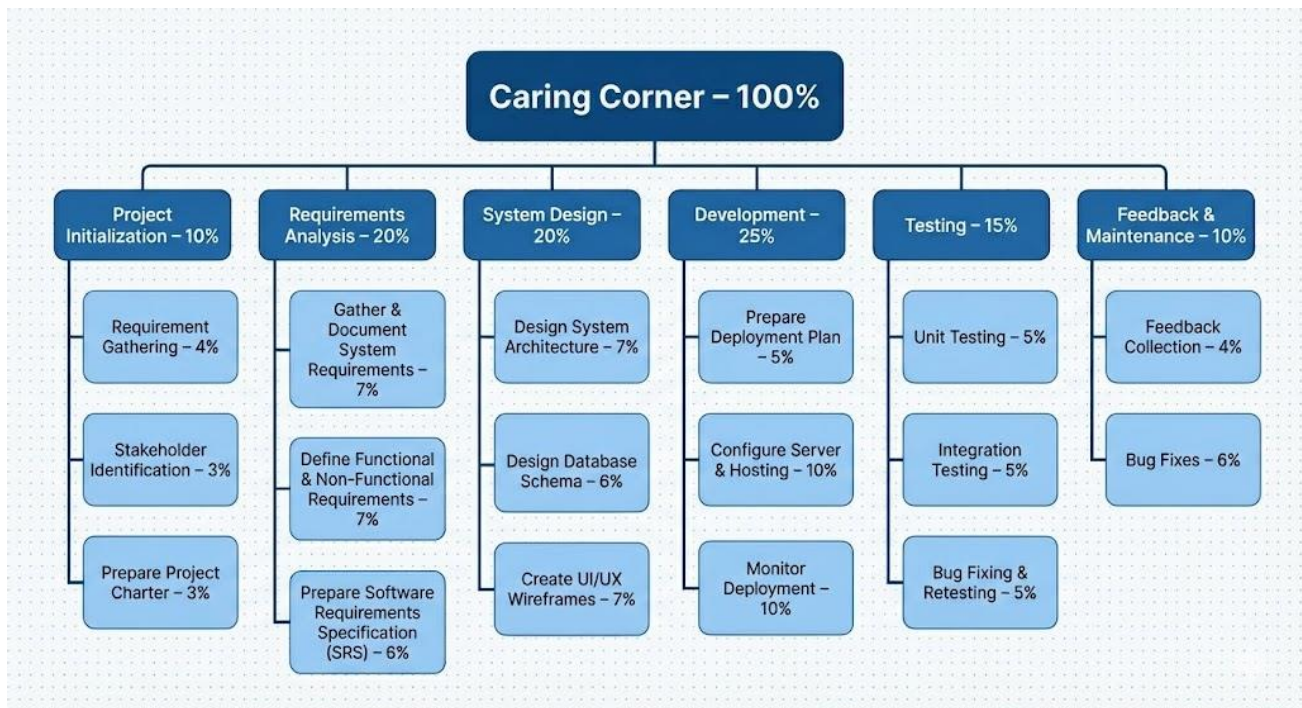
The work is decomposed only to the level necessary for effective planning and execution. Over-decomposition can create unnecessary complexity, while under-decomposition may lead to ambiguity. For Caring Corner, each work package is broken down until it can be realistically estimated, assigned, and tracked within a reasonable timeframe.

5.3 WBS of the Caring Corner Project

The Work Breakdown Structure of the Caring Corner project is designed to fully represent the system's development lifecycle and operational needs.

5.3.1 Compliance with the 100% Rule

The Caring Corner WBS strictly adheres to the 100% Rule. It includes all project activities such as project planning, requirement gathering, system architecture design, UI/UX design, frontend and backend development, payment gateway integration, testing, deployment, feedback handling, and ongoing maintenance. No essential work is omitted, and no out-of-scope activities are included.



5.3.2 Deliverable-Oriented Structure

Each WBS component represents a concrete deliverable. Examples include:

- Online Seat Booking Module
- Admission Management System
- Real-Time Bed Availability Tracking
- Donation System with Payment Integration
- Feedback Collection Module
- Admin Dashboard and Reporting

This structure ensures clarity for both technical team members and non-technical stakeholders.

5.3.3 Mutually Exclusive Design

All deliverables are defined to be independent and non-overlapping. For instance, frontend UI development is separated from backend API development, and testing activities are independent of deployment tasks. This design improves accountability and simplifies progress tracking.

5.3.4 Progressive Decomposition

Major deliverables are broken down into smaller components. For example, the Development phase is decomposed into frontend development, backend development, database integration, and payment gateway configuration. These components are further broken down into manageable tasks such as booking logic implementation, donation transaction handling, and admin permission management.

5.3.5 Consistent Hierarchical Structure

Each branch of the WBS follows a consistent level of detail, ensuring uniformity across different project phases. This consistency improves readability, simplifies estimation, and supports effective project monitoring.

5.3.6 Clear Accountability

Every work package in the Caring Corner WBS has a clearly defined owner. Project planning tasks are assigned to the Project Manager, requirement analysis to the Business Analyst, system design to the System Architect, development tasks to the Development Team, UI/UX components to the UI/UX Designer, and testing activities to the QA team. This clear accountability ensures ownership, reduces confusion, and supports effective project control.

Chapter 6: Risk Management

6.1 What is Risk Management?

Risk Management is a structured and systematic process used to identify, analyze, prioritize, and control potential risks that may negatively impact the success of a project. In software development projects like **Caring Corner – Digital Old Age Home Management System**, risks can arise from technical challenges, management issues, resource limitations, and external dependencies.

The primary objective of risk management in the Caring Corner project is to minimize uncertainties that could affect system reliability, schedule, cost, security, and user trust. By identifying risks early and planning mitigation strategies, the project team can proactively reduce the likelihood and impact of failures, ensuring smooth project execution and successful delivery.

Risk management also improves decision-making, enhances transparency, and ensures preparedness for unexpected events throughout the project lifecycle.

6.2 Risk Identification, Mitigation, and Monitoring

The Caring Corner project involves multiple stakeholders, online financial transactions, real-time data handling, and administrative control. Therefore, effective risk management is essential to maintain system integrity and user confidence.

6.2.1 Risk Mitigation

Risk mitigation refers to the strategies and actions taken to reduce either the probability or impact of identified risks. For the Caring Corner system, mitigation strategies include:

- Using reliable third-party services for payment gateway integration
- Applying validation and error-handling mechanisms
- Performing regular testing and quality assurance
- Implementing secure authentication and authorization controls
- Providing user-friendly interfaces to reduce operational errors

6.2.2 Risk Monitoring

Risk monitoring is an ongoing activity throughout the project lifecycle. The project team continuously reviews identified risks, monitors warning indicators, and evaluates the effectiveness of mitigation plans. Weekly team meetings and progress reviews are used to track risk status and introduce new controls if required.

6.2.3 Risk Management Process

The overall risk management process for Caring Corner follows these steps:

1. Identify potential risks
2. Analyze probability and impact
3. Prioritize risks based on severity
4. Define mitigation strategies
5. Monitor and update risk status regularly

6.3 Risk Plan Sheet

Based on the project scope, system features, and stakeholder needs, the following risks have been identified and prioritized for the Caring Corner project.

Table: Risk Plan Sheet for Caring Corner

Risk ID	Description	Probability	Impact	Priority	Mitigation Plan	Owner
R1	Payment gateway failure	4	5	20	1. Use trusted services like SSLCommerz 2. Add retries	Dev Lead
R2	Admin panel bugs	5	5	25	1. Unit testing and QA reviews	QA Lead
R3	Booking errors	6	3	18	1. Double-check logic 2. Use backups	Dev Team
R4	Low user engagement	2	4	8	1. Add tutorial/demo 2. Simplify UI	PM
R5	Server downtime	2	5	10	1. Use hosting with uptime SLA	Admin

Risk Priority Order: R2 > R1 > R3 > R5 > R4

6.4 Risk Assessment and Prioritization

Risk prioritization in Caring Corner is performed by evaluating both the likelihood of occurrence and the potential impact on the project. High-priority risks such as booking errors and payment failures are addressed early in development due to their direct effect on user trust and system reliability.

Medium and low-priority risks are monitored continuously and mitigated as necessary to prevent escalation.

6.5 Risk Monitoring and Review Plan

To ensure effective risk control, the following monitoring practices are applied:

- Weekly risk review meetings
- Status tracking using shared documents
- Continuous testing during each development sprint
- Immediate escalation of high-impact risks

This approach ensures that risks remain visible, controlled, and manageable throughout the project lifecycle.

6.6 Risk Management Diagram (Conceptual Description)

The risk management workflow for the Caring Corner project follows a continuous cycle:

Risk Identification → Risk Analysis → Risk Prioritization → Mitigation Planning → Risk Monitoring → Review & Update

This cyclical process ensures that risk management is not a one-time activity but an ongoing practice integrated into each sprint and project phase.

Chapter 7: Wideband Delphi Estimation

7.1 What is Wideband Delphi Estimation?

The **Wideband Delphi estimation technique**, originally developed by the RAND Corporation in the 1940s, is a structured and collaborative forecasting approach used to estimate project effort, time, and resources. Over the years, this technique has been widely adopted in project management and software engineering due to its ability to produce accurate and reliable estimates through expert consensus.

Wideband Delphi is particularly valuable because it helps define the **Work Breakdown Structure (WBS)**, clarify assumptions, identify task priorities, and generate realistic effort estimates. Unlike individual estimation methods, it reduces personal bias by encouraging open discussion and constructive feedback among team members. Through multiple rounds of refinement, inaccurate assumptions are corrected, leading to more dependable estimates.

For the **Caring Corner – Digital Old Age Home Management System**, Wideband Delphi is well suited because the project involves multiple interconnected modules such as online seat booking, admission management, real-time bed availability, donation processing, feedback collection, and administrative dashboards. Each module varies in complexity and risk, making collaborative estimation essential for effective planning and execution.

7.2 Wideband Delphi Process

The Wideband Delphi process generally consists of two major phases:

1. An initial phase to generate independent estimates
2. A consensus phase to refine and finalize those estimates

For the Caring Corner project, the process followed these structured steps:

- Choosing the team
- Kickoff meeting
- Individual preparation
- Estimation session
- Assembling tasks
- Reviewing results

Each step contributed to improving estimation accuracy and ensuring shared understanding among team members.

7.2.1 Choosing the Team

Selecting the right estimation team is a critical factor for the success of the Wideband Delphi method. For the Caring Corner project, the following criteria were followed:

1. **Team Selection:**
The project manager selected team members who were directly involved in system design, development, testing, and documentation.
2. **Coordinator Choice:**
A coordinator (moderator) was assigned to manage the estimation process, guide discussions, and document outcomes.
3. **Team Size:**
The estimation group consisted of 3–7 members to ensure effective discussion while maintaining diversity of expertise.
4. **Coordinator's Qualification:**
The coordinator had a clear understanding of the Wideband Delphi process and estimation principles.
5. **Moderator Exclusion:**
To avoid bias, the project manager did not act as the moderator.
6. **Team Collaboration:**
All members committed to providing honest, independent, and collaborative input for each estimation task.

This careful team selection ensured balanced perspectives and credible estimates.

7.2.2 Kickoff Meeting

The kickoff meeting marked the formal start of the estimation process. During this session:

- The moderator explained the Wideband Delphi technique, objectives, and rules to all participants
- The team reviewed the **project vision and scope** of the Caring Corner system
- A high-level task list (approximately 10–20 tasks) was created to represent the top-level WBS
- The **unit of estimation** was agreed upon as *working days*
- Assumptions, constraints, and dependencies were discussed and documented

The facilitator ensured that discussions remained structured and that all participants shared a common understanding of the system requirements before proceeding to estimation.

7.2.3 Individual Preparation

In the individual preparation phase, each team member independently prepared effort estimates for the identified tasks. This phase was conducted without group discussion to avoid influence and group bias.

The individual preparation involved:

1. Creating initial estimates for each task
2. Clearly documenting assumptions used during estimation
3. Excluding project expense-related work, which was treated as overhead
4. Developing a detailed WBS independently
5. Ensuring that all estimates were based on **effort**, not calendar time

This independent approach resulted in a diverse range of estimates, which later helped identify risks, uncertainties, and hidden complexities during group discussions.

7.2.4 Estimation Cost

Task Name	Assigned To	Duration (days)	Expected Cost (\$)
Project Initialization	Saba	5	900
Requirement Documentation	Habib	8	1200
System Design & Architecture	Alom	7	1500
Core System Development	Sameer	20	4000
Testing & Quality Assurance	Habib	6	1300
Deployment Setup	Saba	3	800
Feedback & Maintenance	Alom	3	700
Admin Dashboard Development	Sameer	5	1200
Donation System Integration	Habib	4	1000
Booking & Seat Management	Alom	6	1400
UI/UX Design	Saba	4	1100
Database & ER Implementation	Sameer	5	1600
Security & Access Control	Habib	4	1000
Documentation & User Manual	Saba	3	600
Final Review & Presentation	Alom	2	700
Total	—	85 days	\$19,400

7.2.5 Assembling Tasks

Once consensus was achieved, the team assembled the final task list. Duplicate or unclear tasks were removed, and complex tasks were further decomposed. Best-case, worst-case, and average estimates were calculated to account for uncertainty.

The finalized task list formed the basis for project scheduling, resource allocation, and monitoring.

7.2.6 Reviewing Results

In the final review stage, the team evaluated whether the estimates were realistic and aligned with project constraints, risks, and available resources. After validation, the final estimates were approved and accepted as the baseline for the Caring Corner project.

These estimates play a crucial role in timeline planning, progress tracking, and risk mitigation throughout the project lifecycle.